

CONTRACTOR TO LOCATE AND ASSESS EXISTING
SEWER PUMP STATION PERFORMANCE, CONDITION
AND CAPACITY. CONTRACTOR TO ALLOW ALL COSTS
TO RELOCATE AND REINSTATE SEWER PUMP STATION
AND ALL ASSOCIATED EQUIPMENT AWAY FROM NEW
TRADE'S BUILDING'S FOOTPRINT.

EXISTING SERVICES
SERVICES SHOWN AS EXISTING HAVE BEEN INTERPOLATED FROM RECORD
DRAWINGS. CONTRACTOR TO VERIFY LOCATIONS AND LEVELS PRIOR TO
COMMENCEMENT OF WORK.

INFORMATION REFLECTED ON THIS DRAWING
REPRESENTS HYDRAULIC SERVICES DESIGN
DEVELOPED FOR D&C TENDER ADVICE ONLY, AND
SHOULD NOT BE REGARDED AS COMPLETE. THE D&C
CONTRACTOR SHOULD REFER TO ALL ASSOCIATED
ARCHITECTURAL AND OTHER DISCIPLINE DRAWINGS,
PROJECT BRIEFS AND CLIENT REQUIREMENTS AND
MAKE ADEQUATE ALLOWANCES FOR ALL NECESSARY
HYDRAULIC SERVICES WORKS TO MEET THESE
REQUIREMENTS, EQUIPMENT SCHEDULES AND
COMPLIANCE WITH THE NCC & APPROPRIATE
AUSTRALIAN STANDARDS

A	REVIS	SM	ISSUED FOR D&C TENDER	AA	SM
REV	DATE	BY	DESCRIPTION	CHK	APP

DRAWING STATUS: DESIGN DEVELOPMENT



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CLIENT: CQU

ARCHITECT: THOMSON ADSETT / PEDDLE THORP

PROJECT: CQU ROCKHAMPTON TAFE CONSOLIDATION

TITLE: HYDRAULIC SERVICES
HYDRAULIC SERVICES - PROPOSED SITE PLAN

SCALE @ A1	CHECKED	APPROVED
1:1000	AA	SM
PROJECT NUMBER	DRAWN	DATE
PS229518	SH	JAN 2026

DRAWING NO.	REV
00-HY-DR-300-A00-00-00	A

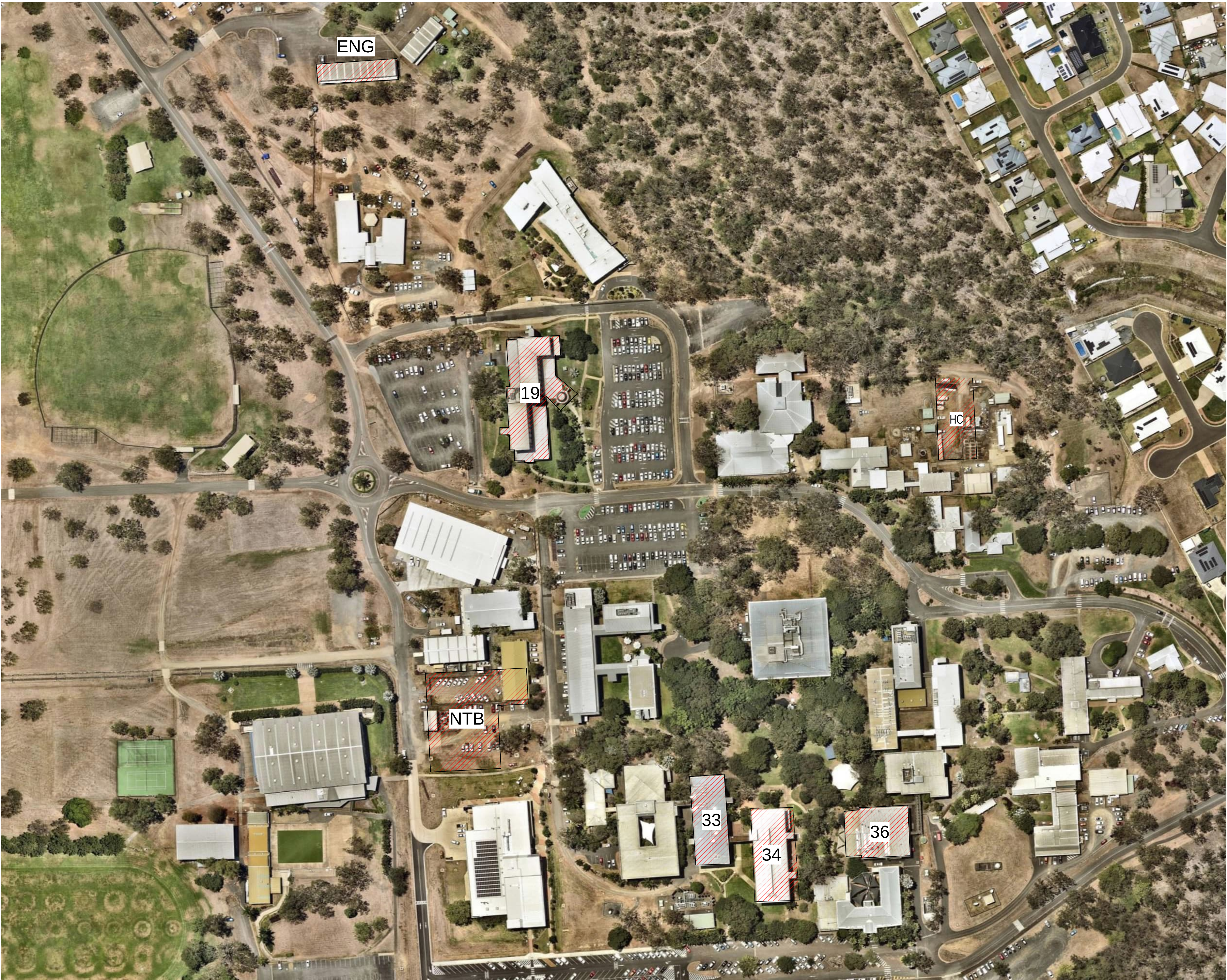
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CQU ROCKHAMPTON TAFE CONSOLIDATION

HYDRAULIC SERVICES

HYDRAULIC DRAWING INDEX

00-HY-DR-000-A00-00-00	COVER SHEET & DRAWING INDEX
00-HY-DR-010-A00-00-00	GENERAL NOTES, LEGENDS & SYMBOLS
00-HY-DR-100-A00-00-01	EQUIPMENT SCHEDULE
00-HY-DR-100-A00-00-02	TYPICAL DETAILS
00-HY-DR-300-A00-00-00	HYDRAULIC SERVICES - PROPOSED SITE PLAN
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B19-HY-DR-520-L03-00-00	BUILDING 19 - LEVEL 03 - WATER LAYOUT
B33-HY-DR-510-L00-00-00	BUILDING 33 - GROUND LEVEL - DRAINAGE LAYOUT
B33-HY-DR-510-L01-00-00	BUILDING 33 - LEVEL 01 - DRAINAGE LAYOUT
B33-HY-DR-510-L02-00-00	BUILDING 33 - LEVEL 02 - DRAINAGE LAYOUT
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B76-HY-DR-510-L00-00-00	BUILDING 36 - LEVEL 01 - WATER SERVICES LAYOUT
B76-HY-DR-510-L00-00-00	HORTICULTURE BUILDING - GROUND LEVEL - DRAINAGE LAYOUT
B76-HY-DR-510-LRF-00-00	HORTICULTURE BUILDING - ROOF LEVEL - DRAINAGE LAYOUT
B76-HY-DR-520-L00-00-00	HORTICULTURE BUILDING - GROUND LEVEL - WATER SERVICES LAYOUT
ENG-HY-DR-510-L00-00-00	ENGINEERING BUILDING - GROUND LEVEL - DRAINAGE LAYOUT
ENG-HY-DR-520-L00-00-00	ENGINEERING BUILDING - GROUND LEVEL - WATER SERVICES LAYOUT
NTB-HY-DR-510-L00-00-00	NEW TRADES BUILDING - GROUND LEVEL - DRAINAGE LAYOUT
NTB-HY-DR-510-L01-00-00	NEW TRADES BUILDING - LEVEL 01 - DRAINAGE LAYOUT
NTB-HY-DR-510-LRF-00-00	NEW TRADES BUILDING - ROOF LEVEL - DRAINAGE LAYOUT
NTB-HY-DR-520-L00-00-00	NEW TRADES BUILDING - GROUND LEVEL - WATER SERVICES LAYOUT
NTB-HY-DR-520-L01-00-00	NEW TRADES BUILDING - LEVEL 01 - WATER SERVICES LAYOUT



LOCALITY PLAN
N.T.S.

INFORMATION REFLECTED ON THIS DRAWING REPRESENTS HYDRAULIC SERVICES DESIGN DEVELOPED FOR D&C TENDER ADVISE ONLY, AND SHOULD NOT BE REGARDED AS COMPLETE. THE D&C CONTRACTOR SHOULD REFER TO ALL ASSOCIATED ARCHITECTURAL AND OTHER DISCIPLINE DRAWINGS, PROJECT BRIEFS AND CLIENT REQUIREMENTS AND MAKE ADEQUATE ALLOWANCES FOR ALL NECESSARY HYDRAULIC SERVICES WORKS TO MEET THESE REQUIREMENTS, EQUIPMENT SCHEDULES AND COMPLIANCE WITH THE NCC & APPROPRIATE AUSTRALIAN STANDARDS

B	12.02.2026	SH	ISSUED FOR D&C TENDER	AA	SM
A	10.02.2026	SH	ISSUED FOR D&C TENDER	AA	SM
REV	DATE	BY	DESCRIPTION	CHK	APP

DRAWING STATUS
DESIGN DEVELOPMENT



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TITLE: HYDRAULIC SERVICES
COVER SHEET & DRAWING INDEX

SCALE @ A0	N.T.S.	CHECKED	AA	APPROVED	SM
PROJECT NUMBER	PS229518	DRAWN	SH	DATE	JAN 2026

DRAWING NO.	00-HY-DR-000-A00-00-00	REV.	B
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STORMWATER GRATED OUTLETS			
ITEM	MANUFACTURER	MODEL	SIZE
GO(1)	SPS	PUSH IN FLOOR DRAIN	130 x 100
GO(2)	SPS	TRUFLOW FLOOR DRAIN*	260 x 100
BDO	SPS	STANDARD VERTICAL	100 X 50
PDO	SPS	STANDARD VERTICAL	260 X 100 X 100
SDO	SPS	ROUND PUSH IN FLOOR DRAIN	100 X 50
* FINAL MODEL SELECTION WITHIN PRODUCT RANGE, DEPENDANT ON APPLICATION (TOPPING SLAB / NO TOPPING SLAB)			

SCHEDULE OF INSULATION			
TYPE	LOCATION	SPECIFICATION	DESCRIPTION
THERMAL	HOT WATER CIRCULATION PIPEWORK & DEADLEGS	THERMOTEC 4 ZERO	25mm WALL THICKNESS CLOSED CELL POLYETHYLENE
THERMAL	EXTERNAL HOT WATER CIRCULATION PIPEWORK	THERMOTEC 4 ZERO	25mm WALL THICKNESS CLOSED CELL POLYETHYLENE WITH 0.5mm THICK METALLIC COATED STEEL SHEATHING
ACOUSTIC	ALL SANITARY AND STORMWATER DRAINAGE ABOVE HABITABLE/RETAIL AREAS	SOUNDLAG 4525C	ACHIEVE MINIMUM SOUND LEVELS SPECIFIED IN ACOUSTIC REPORT
CONDENSATE LAGGING	MECHANICAL TUNDISH TO FLOOR WASTE (INCLUSIVE) OR TRAPPED TUNDISH TO DOWNSTREAM BRANCH OF SANITARY PLUMBING	ARMAFLEX	19mm WALL THICKNESS CLOSED CELLED ELASTOMER

EQUIPMENT SCHEDULE - TREATMENT DEVICES AND PLANT						
ITEM	EQUIP TAG	MAKE	MODEL	CAPACITY (L/HR)	DIMENSIONS (mm)	COMMENTS
GREASE INTERCEPTOR TRAP	GIT(1)	HALGAN	MGTS 5000	5000	1940Hx1625Wx320D	PE GREASE TRAP IN GROUND WITH CLASS 'D' ACCESS LIDS (AND RISERS AS REQUIRED) FINISHED 25mm AFSL
OLI/WATER SEPARATOR	OWS(2)	HALGAN	OWHS 5000	5000	1940Hx1625Wx320D	PE IN GROUND WITH CLASS 'D' ACCESS LIDS (AND RISERS AS REQUIRED) FINISHED 25mm AFSL, 1100 INLET & OUTLET CONNECTIONS WITH OUTLET TO BE TRAPPED.
NEUTRALISING BOX	NB (1)	SCIENCE 2 MEDICAL	50 LITRE	50	455Hx320Wx485L	PVC BELOW BENCH INSTALLED TO MANUFACTURER'S RECOMMENDATIONS AND LOCAT AUTHORITY REQUIREMENTS.

VALVE SCHEDULE				
ITEM	MANUFACTURER	MODEL	SIZE	DESCRIPTION
INDIVIDUAL FIXTURE ISOLATION VALVE	ARCO	MIN STOP	15	QUARTER TURN CHROME PLATED 15mm MINI ISOLATION STOP
AUTOMATIC URINAL FLUSHING DEVICE	Zip	FLUSHMASTER 4091	NA	COMPLETE WITH OPTIONAL AIR BREAK FOR DIRECT INJECTION
GAS EMERGENCY SHUT OFF VALVE	SYSTEM CONTROL ENGINEERING	GASGUARD		CONTROL PANEL WIRED TO FIP
HT(1)	NA	NA		HOSE TAP WITH VANDAL PROOF HANDLE AND KEY
HT(2)	NA	NA	15	15mm CHROME PLATED
IV	NA	NA	15-65	QUARTER TURN BALL TYPE WITH BRASS BODY AND STAINLESS STEEL BALL COMPLETE WITH DOWNSTREAM UNION
IV	NA	NA	80-200	BRASS BODY FLANGED GATE VALVE OR BUTTERFLY VALVE
SV(1)	DOBBIE	NA	50-200	PN16 FLANGED RESILIENT SEATED GATE VALVE SUITABLE FOR IN GROUND INSTALLATION WITH ACCESS RISER TO SURFACE AND 'WM' (WATER MAIN) MARKED CAST IRON COVER AS PER DETAIL
SV(2)	CIMBERIO	CIM 40612CR	20-40	THREADED BRASS BALL VALVE WITH KEYS SQUARE HEAD SUITABLE FOR IN GROUND INSTALLATION WITH ACCESS RISER TO SURFACE AND 'WM' (WATER MAIN) MARKED CAST IRON COVER AS PER DETAIL
GIV	ZETCO	SERIES 1130	15-100	FULL FLOW AGA APPROVED GAS BALL VALVE
TMV(1)	ENWARE	AQUABLEND 1000 (ATM710)	15mm INLET WITH 1500mm OUTLET	LEAD FREE THERMOSTATIC MIXING VALVE - SET AT TEMPERATURE NOMINATED ON SCHEMATICS
TMV(2)	ENWARE	AQUABLEND 1000 (ATM5710H-350)	15mm INLET WITH 1500mm OUTLET	LEAD FREE THERMOSTATIC MIXING VALVE IN 350 x 350mm STAINLESS STEEL WALL BOX, HWIC/WWP PIPES AT BOTTOM, WITH SECURE HEAVY DUTY LID - SET AT TEMPERATURE NOMINATED ON SCHEMATICS
TMV(3)	ENWARE	AQUABLEND 1000 (ATM712)	20mm INLET WITH 2025mm OUTLET	LEAD FREE THERMOSTATIC MIXING VALVE - SET AT TEMPERATURE NOMINATED ON SCHEMATICS
TMV(4)	ENWARE	AQUABLEND 1000 (ATM5712H-350)	20mm INLET WITH 20mm OUTLET	LEAD FREE THERMOSTATIC MIXING VALVE IN 350 x 350mm STAINLESS STEEL WALL BOX, HWIC/WWP PIPES AT BOTTOM, WITH SECURE HEAVY DUTY LID - SET AT TEMPERATURE NOMINATED ON SCHEMATICS
TV(1)	RMC			SIZED TO SUIT PIPE SIZE NOMINATED ON DRAWINGS, COMPLETE WITH SS WALL BOX AND INTEGRATED HINGED COVER. TO BE SET @ 50°C UNLESS ADVISED OTHERWISE IN SCHEMATIC
TV(2)	RMC			SIZED TO SUIT PIPE SIZE NOMINATED ON DRAWINGS. TO BE SET @ 50°C UNLESS ADVISED OTHERWISE IN SCHEMATIC
PRV	BERMAD	SERIES 700	50-200	50mm & LARGER, SET @ 500kPa PRESSURE NOMINATED ON DRAWINGS & WITH MATCHING LOW FLOW BYPASS PRV ASSEMBLY
RP2D(1)	TYCO	RPB3CO	15-50	BACKFLOW PREVENTION DEVICE COMPLETE WITH UIS AND DIS ISOLATION VALVES, STRAINER, BARREL UNIONS
DCV	TYCO	DCB3CO	15-50	BACKFLOW PREVENTION DEVICE COMPLETE WITH UIS AND DIS ISOLATION VALVES, STRAINER, BARREL UNIONS
ASD	AQUASAVER			AUTOMATIC SWITCHING DEVICE - HYDRAULIC OPERATED WITH 1" BSP INLET AND OUTLETS
BV	CIMBERIO	FIXED ORIFICE		FIXED ORIFICE TYPE BALANCING VALVE COMMISSIONED TO REQUIRED FLOW RATE
TPD	MAG-MIFAB	80102-MRA-500		PRESSURE DROP AUTOMATED PRIMING VALVE, PRIMES 1-10 TRAPS
AAV	STUDOR	MAXI-VENT or AAV		ONE WAY AIR VALVE, FLOW RATE AND SIZE IN ACCORDANCE WITH AS/NZS3500.2
RV	I-PLEX	1588.100	100	PVC IN GROUND REFLUX VALVE WITH 1500 RISER TO SURFACE TO PERMIT ACCESS TO VALVE ASSEMBLY

EQUIPMENT SCHEDULE - WATER TANKS						
ITEM	EQUIP TAG	MAKE	MODEL	EFFECTIVE STORAGE CAPACITY (LITRES)	CONSTRUCTION	FEATURES AND ALLOWANCES
RAIN WATER HARVESTING TANK	RWHT(1)	BUSHMAN TANKS	AQUALINE STEEL LINER	30,000	GALVANISED STEEL	4.32m (Ø) x 2.35m(H) ABOVE GROUND - OVERFLOW TO ATMOSPHERE - EXTERNAL ISOLATION VALVES - LOW LEVEL SLUDGE DRAINS - LEVEL SENSING (SUCTION PUMP STOP AND MAINS DIVERSION ACTUATION) - INLET SCREENS AS PER DOCUMENTATION
RAIN WATER HARVESTING TANK	RWHT(2)	BUSHMAN TANKS	AQUALINE STEEL LINER	15,000 EACH FINAL SIZE TBC	GALVANISED STEEL	3.0m (Ø) x 2.3m(H) ABOVE GROUND - 2 x TANKS INTERCONNECTED WITH TOTAL STORAGE CAPACITY 30KL - INTERCONNECTING PIPEWORK WITH INDIVIDUAL TANK ISOLATING AND SINGLE PUMP SUCTION - LOW LEVEL SLUDGE DRAINS - OVERFLOWS TO ATMOSPHERE - INLET SCREENS AS PER DOCUMENTATION
TRADE WASTE HOLDING TANK	HOL(1)	TBC BY CONTRACTOR	TBC BY CONTRACTOR	1000 (EFFECTIVE)	PRE-CAST CONCRETE	1.05m (Ø) x 1.5m(H) BELOW GROUND - EXAGGERATED FALL TOWARDS DIAPHRAGM PUMP SCTION POINT - 1000 CHAMBER VENT - 1100 INLET - 300 (MIN) SUCTION OUTLET - CLASS 'D' TRAFFICABLE LID AND FRAME FINISHED 25mm AFSL - RISERS TO FINISHED SURFACE LEVEL AS REQUIRED. - CONTROL PANEL - HIGH LEVEL ALARM FLOAT SWITCH - ANTI-FLOATATION BASE

EQUIPMENT SCHEDULE - HOT WATER PLANT & BOILING/CHILLED WATER UNITS								
ITEM	EQUIP TAG	MAKE	MODEL	TYPE	BOOSTERS/TANKS	FLUE SYSTEM	ROOM VENTILATION (NATURAL/MECHANICAL)	ELEC REQ (kW)
HOT WATER PLANT	HWP(1)	RHEEM	COMMERCIAL HEAT PUMP	ELECTRIC BOOST HEAT PUMP PLANT: -1 x 95301500 A2W HP 15kW -1 x 61043050 410 LITRE PREHEAT WATER STORAGE TANKS -1 x 61333507 315 LITRE HEAVY DUTY ELECTRIC BOOST STORAGE TANKS EACH WITH 3x3.6kW ELEMENTS -1x PRIMARY PUMP (REFER PUMP SCHEDULE) -1 x HOT WATER DUAL CIRCULATION PUMP SET (REFER PUMP SCHEDULE)	5 / 2	NA	NA	415V, 9.84kW, 40AMPS PER PHASE
HOT WATER UNIT	HWU(1)	RHEEM	613-315	315 LTR ELECTRIC STORAGE SET @ 60°C LOCATED WITHIN CLPBOARD, COMPLETE WITH IV, NON RETURN VALVE, EXPANSION CONTROL VALVE, PRESSURE AND TEMPERATURE RELIEF VALVE. HWU SHALL BE PLACED IN AN ALL IN ONE SAFETRAY EQUAL TO 'UNTRAY' WHICH SHALL INCORPORATE A SAFETRAY, ELEVATING POSTS, INTEGRATED SAFETRAY DRAIN AND TUNDISH. VALVE SHALL DISCHARGE OVER TUNDISH.	0 / 1	NA	NA	415V, 10.8kW
HOT WATER UNIT	HWU(2)	RHEEM	492-315	315 LTR ELECTRIC STORAGE SET @ 60°C LOCATED WITHIN PLANT ROOM, COMPLETE WITH IV, NON RETURN VALVE, EXPANSION CONTROL VALVE, PRESSURE AND TEMPERATURE RELIEF VALVE. HWU SHALL BE PLACED IN AN ALL IN ONE SAFETRAY EQUAL TO 'UNTRAY' WHICH SHALL INCORPORATE A SAFETRAY, ELEVATING POSTS, INTEGRATED SAFETRAY DRAIN AND TUNDISH. VALVE SHALL DISCHARGE OVER TUNDISH.	0 / 1	NA	NA	415V, 7.2kW
HOT WATER UNIT	HWU(3)	RHEEM	613-050	50 LTR ELECTRIC STORAGE SET @ 60°C LOCATED WITHIN PLANT ROOM, COMPLETE WITH IV, NON RETURN VALVE, EXPANSION CONTROL VALVE, PRESSURE AND TEMPERATURE RELIEF VALVE. HWU SHALL BE PLACED IN AN ALL IN ONE SAFETRAY EQUAL TO 'UNTRAY' WHICH SHALL INCORPORATE A SAFETRAY, ELEVATING POSTS, INTEGRATED SAFETRAY DRAIN AND TUNDISH. VALVE SHALL DISCHARGE OVER TUNDISH.	0 / 1	NA	NA	415V, 10.8kW
HOT WATER UNIT	HWU(4)	STIEBEL ELTRON	DHE 27 AU	INSTANTANEOUS ELECTRIC (60°C)	1 / 0	NA	NA	415V, 28.1kW, 40AMPS PER PHASE
HOT WATER UNIT	HWU(5)	STIEBEL ELTRON	DEL 27 PLUS	INSTANTANEOUS ELECTRIC (50°C)	1 / 0	NA	NA	415V, 28kW, 39AMPS PER PHASE
HOT WATER UNIT	HWU(6)	STIEBEL ELTRON	DEL 18 PLUS	INSTANTANEOUS ELECTRIC (50°C)	1 / 0	NA	NA	240V, 15.4kW, 27AMPS PER PHASE
HOT WATER UNIT	HWU(7)	STIEBEL ELTRON	DHB-E 13 LCD	INSTANTANEOUS ELECTRIC (60°C)	1 / 0	NA	NA	240V, 14.5kW, 20AMPS PER PHASE
BOILING/CHILLED WATER UNIT	BCWU(1)	ZIP	GS BC 240175	BOILING/CHILLED DISPENSING FONT STYLE & COLOUR SELECTION; AND SINK MIXER STYLE & COLOUR SELECTION TO ARCHITECTURAL SPECIFICATION.	NA	NA	NA	2 x 10A GPO
BOILING/CHILLED/HOT WATER UNIT	BCHWU(1)	ZIP	GS BCWA 240175	BOILING/CHILLED/HOT/AMBIENT DISPENSING FONT STYLE & COLOUR SELECTION; AND SINK MIXER STYLE & COLOUR SELECTION TO ARCHITECTURAL SPECIFICATION.	NA	NA	NA	2 x 10A GPO

FLOOR DRAINS			
ITEM	MANUFACTURER	MODEL	SIZE
FW	*	*	*
BTGP	FOX ENVIRONMENTAL	PT600100	600 x 600
FWP(1)	SPS	PUSH IN FLOOR DRAIN WITH "GRATE SEAL" ONE WAY FLAP VALVE	130 x 100
BFW	SPS	VAN LEVEL BUCKET TRAP R150SA2 OR PUSH IN BUCKET TRAP R150SR-8T	150 x 100
BFW	SPS	150mm ROUND PUSH IN BUCKET TRAP	150 x 100
SHR	*	*	*
TTD	NA	PVC LIT & 100/65 TRAP BOTTOM	100 x 65
TTD(TD1)	NA	PVC LIT	100 x 50
TTD(TD2)	NA	PVC LIT	150 x 100
TTD(TD3)	NA	HDPE TAPERED REDUCER	75 x 56
TTD(TD4)	NA	HDPE TAPERED REDUCER	160 x 110
TTD(TD5)	M.A. GRIFFITH	78000	WALL BOX COMPLETE WITH SIGHT GLASS
* REFER ARCHITECTURAL CONSULTANT DRAWINGS			

INSPECTION OPENINGS				
ITEM	MANUFACTURER	MODEL	SIZE	DESCRIPTION
IOS	REECE	202680	100	TO ALL INTERNAL IOS - POLISHED BRASS FINISHED FLOOR CLEAROUT TO FINISHED FLOOR LEVEL
IOS	REECE	202660	100	TO ALL INTERNAL IOS - CHROME FINISHED BRASS FLOOR CLEAROUT TO FINISHED FLOOR LEVEL
IOS	SPS	R20SSCRN	150	NON SLIP 304 STAINLESS STEEL BOLTED TRAP SCREW

SCHEDULE OF ARTICULATED DRAINAGE FITTINGS						
FITTING	MANUFACTURER	MODEL	SIZE	MOVEMENT RANGE	SWIVEL ANGLE	EXPANDED LENGTH
ARTICULATION	STORMPLASTICS	SJ110015	100	50mm	+/- 15°	325mm
ARTICULATION	STORMPLASTICS	SJ115015	150	60mm	+/- 15°	375mm
ARTICULATION	STORMPLASTICS	SJ22520	225	130mm	+/- 15°	520mm
EXPANSION	STORMPLASTICS	DW101050	100	150mm	NA	470mm
EXPANSION	STORMPLASTICS	DW150240	150	240mm	NA	790mm
EXPANSION	STORMPLASTICS	DWS225150	225	150mm	NA	270mm
COMB EXPARTIC	STORMPLASTICS	SJ100EXP/COMBOFF	100	100mm	+/- 15°	325mm
COMB EXPARTIC (HDPE)	PLASTEC	16675	110	80mm	40°	231mm

STORMWATER PITS AND TRENCH GRATES					
ITEM	MANUFACTURER	MODEL	SIZE	DESCRIPTION	COVER TYPE
SWGPI(1)	-	-	300mm x 300mm x 450mm (DEEP)	POLYCRETE	C
SWGPI(2)	-	-	600mm x 600mm x 600mm (DEEP)	POLYCRETE	C
SWGPI(3)	ACO DRAIN	TYPE 66H	600mm x 600mm x 900mm (DEEP)	POLYCRETE	C
SWGPI(4)	-	-	450mm x 450mm x 600mm (DEEP)	POLYCRETE	C
GTD	ACO DRAIN	K200	200mm WIDE x 200-400mm DEEP (INTERNAL)	CHANNEL CLASS 'D'	B

SCHEDULE OF PIPEWORK			
SERVICE	LOCATION	MATERIAL	NOMINAL PIPE SIZE
STORMWATER	ABOVE & BELOW GROUND	PVC - 'DWV' GRADE	100-300
SANITARY DRAINAGE	ABOVE & BELOW GROUND	PVC - 'DWV' GRADE	40-300
SANITARY DRAINAGE (HIGH TEMP DISCHARGE)	ABOVE & BELOW GROUND	HDPE or PPR	40-300
SANITARY PLUMBING	ABOVE GROUND	PVC - 'DWV' GRADE	40-300
VENT PIPES	ABOVE GROUND	PVC - 'DWV' GRADE	40-150
SANITARY RISING MAIN	ABOVE & BELOW GROUND	PE100 PN16 or PVC-U PRESSURE PN12	25-150
SUB SOIL DRAINAGE PLANTER	BELOW GROUND	PE SQUARE DRAINAGE CELL WITH GEOTEXTILE FABRIC WRAP - ATLANTIS	NA
SUB SOIL DRAINAGE	BELOW GROUND	HDPE SLOTTED CORRUGATED DRAINAGE PIPE WITH RTA FILTER SOCK - 'DRAINOLK' BY VINDEX	110-160
TRADE WASTE DRAINAGE	ABOVE & BELOW GROUND	HDPE WITH FUSION WELD COUPLINGS or PPR	110-160
TRADE WASTE STACK	ABOVE & BELOW GROUND	HDPE WITH FUSION WELD COUPLINGS or PPR	110-160
TRADE WASTE VENT	ABOVE & BELOW GROUND	PVC - 'DWV' GRADE	50-100
CONDENSATE DRAINAGE	ABOVE GROUND	PVC - 'DWV' GRADE (LAGGED TO PREVENT EXTERNAL PIPE CONDENSATION)	25-150
COLDHOTRAIN/RECYLED WATER	ABOVE GROUND	COPPER TUBES TO AS1432 - TYPE 'A' & TYPE 'B' ACCORDING TO PRESSURE REQUIREMENTS	15-200
COLDHOTRAIN/RECYLED WATER	BELOW GROUND	COPPER TUBES TO AS1432 - TYPE 'A' & TYPE 'B' (ACCORDING TO PRESSURE REQUIREMENTS) WRAPPED WITH PETROLEUM IMPREGNATED TAPE or PVC-PRESSURE PN20 or PE100 PN20	15-200
COLDHOTRAIN/RECYLED WATER - HIGH PRESSURE	ABOVE GROUND	STAINLESS STEEL 316 - SCHEDULE 10S - 'PROCHEM' ROLL GROOVE	50-300
COLDHOTRAIN/RECYLED WATER - ROUGH INS	ABOVE GROUND	COPPER TUBES TO AS1432 - TYPE 'B' or CROSS-LINKED POLYETHYLENE (PEX) - 'REHAU'	15-25
GAS SERVICE	ABOVE GROUND	COPPER TUBES TO AS1432 - TYPE 'B'	15-200
GAS SERVICE	BELOW GROUND	COPPER TUBES TO AS1432 - TYPE 'B' WRAPPED WITH PETROLEUM IMPREGNATED TAPE	15-200

EQUIPMENT SCHEDULE - WATER FILTRATION						
ITEM	EQUIP TAG	MAKE	MODEL	SCREEN (MICRON)	SYSTEM	No OFF
RAINWATER HARVESTING FILTERS	RWF(1)	SOUTHLAND FILTRATION	DEFENDER D7	100 AND 5	PACKAGED DUAL STAGED FILTRATION WITH AUTOMATED BACKWASH AND UV DISINFECTION	1
RAINWATER HARVESTING FILTERS	RWF(2)	TBC	TBC	TBC	DUAL STAGED FILTRATION WITH AUTOMATED BACKWASH AND UV DISINFECTION	TBC
ICE MACHINE FILTERS	IMF	SOUTHLAND FILTRATION	ROBUST 60 L HR	NA	60 LHR CAPACITY DIRECT FLOW REVERSE OSMOSIS PACKAGED UNIT	1 PER ICE MACHINE (REFER ARCHITECTURAL INTERIOR FFE SCHEDULE FOR QUANTITY)

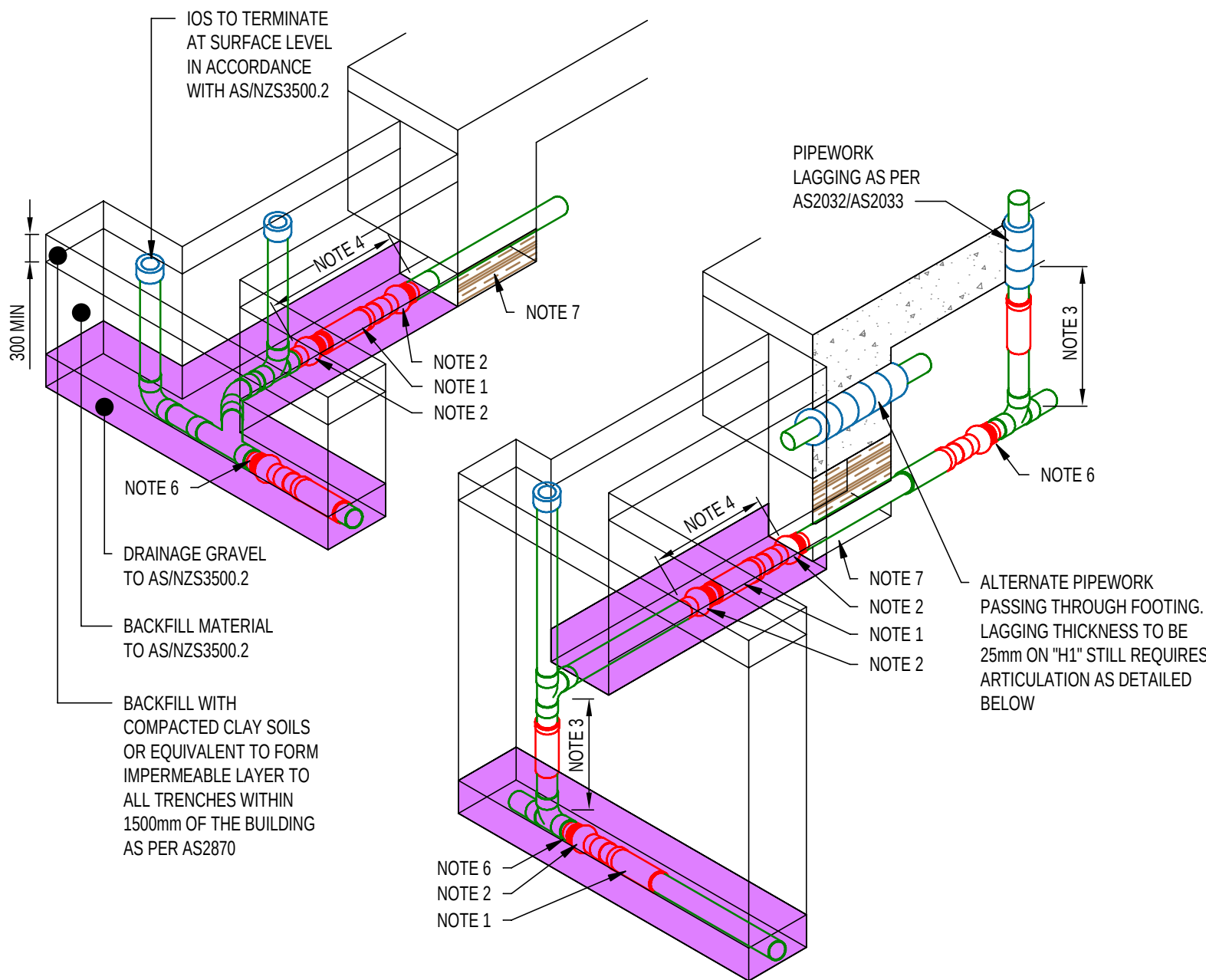
EQUIPMENT SCHEDULE - PUMPS AND PUMP STATIONS						
ITEM	EQUIP TAG	MAKE/MODEL	DESCRIPTION	FLOW PER PUMP (L/S)	TOTAL FLOW (L/S)	ESSENTIAL POWER
SANITARY DRAINAGE PUMP STATION	SDPS	TBC BY CONTRACTOR	2.1m (Ø) DIAMETER PROPRIETARY PACKAGED PUMP STATION WITH THE FOLLOWING FEATURES: - DUAL ACCESS LIDS (APPROPRIATELY SIZED) TO BE CLASS RATED ACCORDING TO THE APPLICABLE EXPECTED TRAFFIC LOAD. - AUTO COUPLING DUAL GRINDER CUTTER SUBMERSIBLE PUMPS, WITH STAINLESS STEEL FIXED GUIDE RAILS AND SUPPORTS. - ALTERNATING DUTY/STANDBY MODES - ULTRASONIC OR TRANSDUCER LEVEL SENSORS. - INTEGRATED VALVE BOX WITH ASSOCIATED VALVES AS PER DETAIL. - TO INCLUDE AN EMERGENCY KEY STORAGE VOLUME OF 345 LITRES - CONTROL PANEL FEATURING: - IP66 RATED - CABINET DOOR REED SWITCH - AUTO PUMP CONTROL WITH ALTERNATING DUTY AND STANDBY - VISUAL FAULT INDICATION STROBE LIGHT - HOUR METER FOR EACH PUMP - MANUAL PUMP CONTROL FOR EACH PUMP - BMS OUTPUTS FOR: - EACH PUMP RUN - EACH PUMP FAULT - HIGH LEVEL ALERT	3	6	35
HOT WATER PRIMARY PUMP	HWPP(1)	GRUNDFOS CM3-2	HEAT PUMP PRIMARY PUMP SET COMPLETE WITH CONTROL PANEL.	-	-	NO
HOT WATER CIRCULATION PUMPS	HWCP(1)	GRUNDFOS UPS 20-60N	PACKAGED DUAL VARIABLE SPEED HOT WATER CIRCULATION PUMP SET COMPLETE WITH: - CONTROL PANEL - ALTERNATING DUTY/STANDBY PUMPS - INCOMING FLOW GAUGE - CONTROL PANEL TO HAVE INDIVIDUAL PUMP CONTROL (AUTOMANUAL/OFF), PUMP HOUR METERS, BMS OUTPUTS FOR EACH PUMP RUN, EACH PUMP FAULT - PUMP SPEED TO ENSURE MAX VELOCITY OF 1m/s.	0.077	0.077	4.3
HOT WATER CIRCULATION PUMPS	HWCP(2)	GRUNDFOS UPS 20-60N	PACKAGED DUAL VARIABLE SPEED HOT WATER CIRCULATION PUMP SET COMPLETE WITH: - CONTROL PANEL - ALTERNATING DUTY/STANDBY PUMPS - INCOMING FLOW GAUGE - CONTROL PANEL TO HAVE INDIVIDUAL PUMP CONTROL (AUTOMANUAL/OFF), PUMP HOUR METERS, BMS OUTPUTS FOR EACH PUMP RUN, EACH PUMP FAULT - PUMP SPEED TO ENSURE MAX VELOCITY OF 1m/s.	0.05	0.1	2.2
RAINWATER PUMP	RWP(1)	DAVEY KRB4 (H490) 13	COMPLETE WITH MAINS BACK-UP DIVERSION VALVE AND FLOAT SWITCH.	1	1	40
RAINWATER PUMP	RWP(2)	TBD	PENDING HORTICULTURE BUILDING IRRIGATION REQUIREMENTS	TBC	TBC	NO

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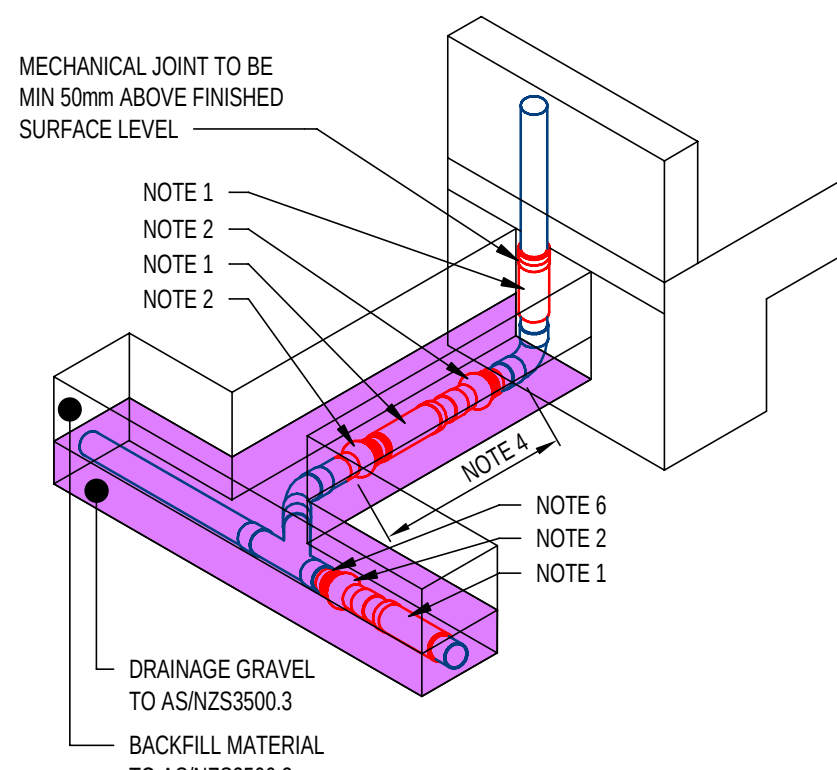
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A	19.02.2026	SH	ISSUED FOR D&C REVIEW	AA	SM
REV	DATE	BY	DESCRIPTION	CHK	APP
DRAWING STATUS					
DESIGN DEVELOPMENT					
					
900 Ann Street, Level 12, Fortitude Valley 4006 Tel: +61-7-3854-6200 Fax: +61-7-3854-6500 http://www.wsp.com					
CLIENT: CQU					
ARCHITECT: THOMSON ADSETT / PEDDLE THORP					
PROJECT: CQU ROCKHAMPTON TAFE CONSOLIDATION					
TITLE: HYDRAULIC SERVICES EQUIPMENT SCHEDULE					
SCALE & NO: N.T.S					
CHECKED: AA		APPROVED: SM		PROJECT NUMBER: PS229518	
DRAWN: SH		DATE: JAN 2026		DRAWING NO: 00-HY-DR-100-A00-00-01	
REV: B					© WSP Australia Pty Ltd

AREA OF WORKS HAS A SITE SOIL CLASSIFICATION OF "H1"
IN ACCORDANCE WITH AS2870. FOR FURTHER INFORMATION
REFER TO GEOTECHNICAL REPORT (BY OTHERS)

- NOTES
1. EXPANSION JOINT WRAPPED IN A SUITABLE MATERIAL TO ALLOW FULL MOVEMENT AND TO PREVENT THE INGRESS OF FOREIGN MATTER. MINIMUM TRAVEL: "H1" 50mm
 2. ARTICULATION JOINT WRAPPED IN A SUITABLE MATERIAL TO ALLOW FULL MOVEMENT AND TO PREVENT THE INGRESS OF FOREIGN MATTER. INSTALLED TO MANUFACTURERS SPECIFICATIONS.
 3. IF VERTICAL DROP GREATER THAN 100mm THEN EXPANSION JOINT TO BE INSTALLED.
 4. PIPEWORK GRADED AT MINIMUM 1:60 FOR A MINIMUM DISTANCE OF 300mm DOWNSTREAM OF EXITING STRUCTURAL SLAB.
 5. ALL TRENCHING TO BE GRADED UNIFORMLY WITH PIPEWORK TO ENSURE TRENCH DRAINAGE.
 6. EXPANSION AND SWIVEL JOINT DOWNSTREAM OF EVERY JUNCTION (TYPICAL) & CHANGE IN DIRECTION (>45°). FOR H1 SITES, A COMBINATION SWIVEL/EXPANSION JOINT MAY BE UTILISED.
 7. COMPACTED CLAY SOILS OR EQUIVALENT IN TRENCH TO FORM IMPERMEABLE "PLUG" TO PREVENT THE INGRESS OF WATER THROUGH TRENCHING AS PER AS2870.
 8. THIS ARTICULATION DESIGN HAS BEEN PREPARED IN CONJUNCTION WITH MANUFACTURER AND INDUSTRY INFORMATION, TO SATISFY THE REQUIREMENTS OF THE PLUMBING AND DRAINAGE REGULATION 2013 SCHEDULE 6.
 9. THIS DETAIL SHALL BE FULLY COMPLIED WITH IN REGARDS TO CONTRACTOR INSTALLATION, UNLESS WRITTEN APPROVAL IS PROVIDED BY WSP CERTIFYING ENGINEER OR THEIR APPROVED REPRESENTATIVE.

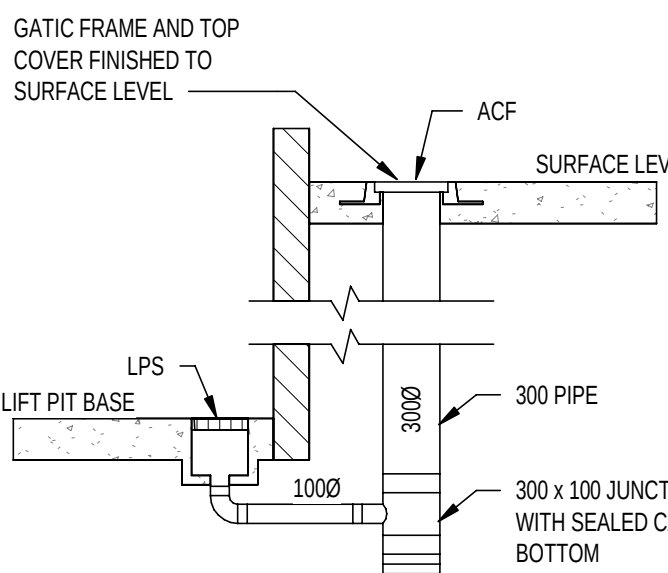


1 IN GROUND DRAINAGE (STORMWATER & SANITARY)
N.T.S.

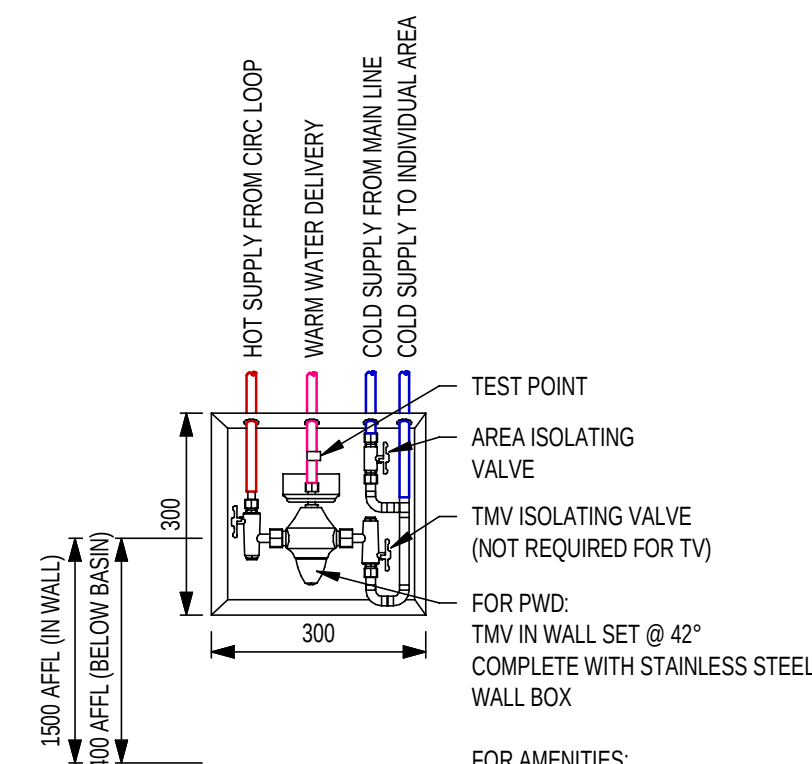


2 STORMWATER DOWNPIPES
N.T.S.

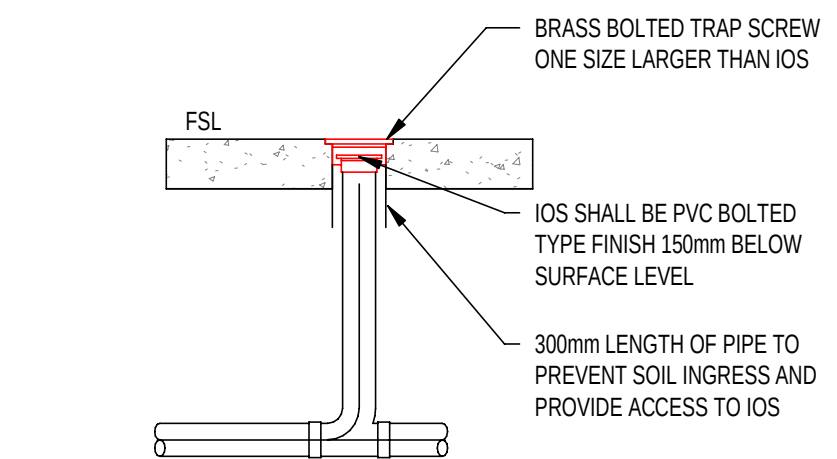
FLEXIBLE DRAINAGE DETAIL
N.T.S.



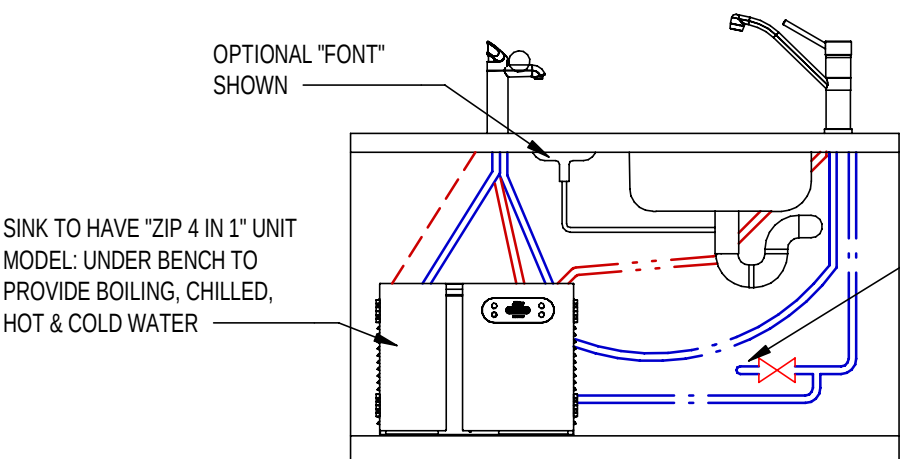
LIFT SUMP DRAINAGE
N.T.S.



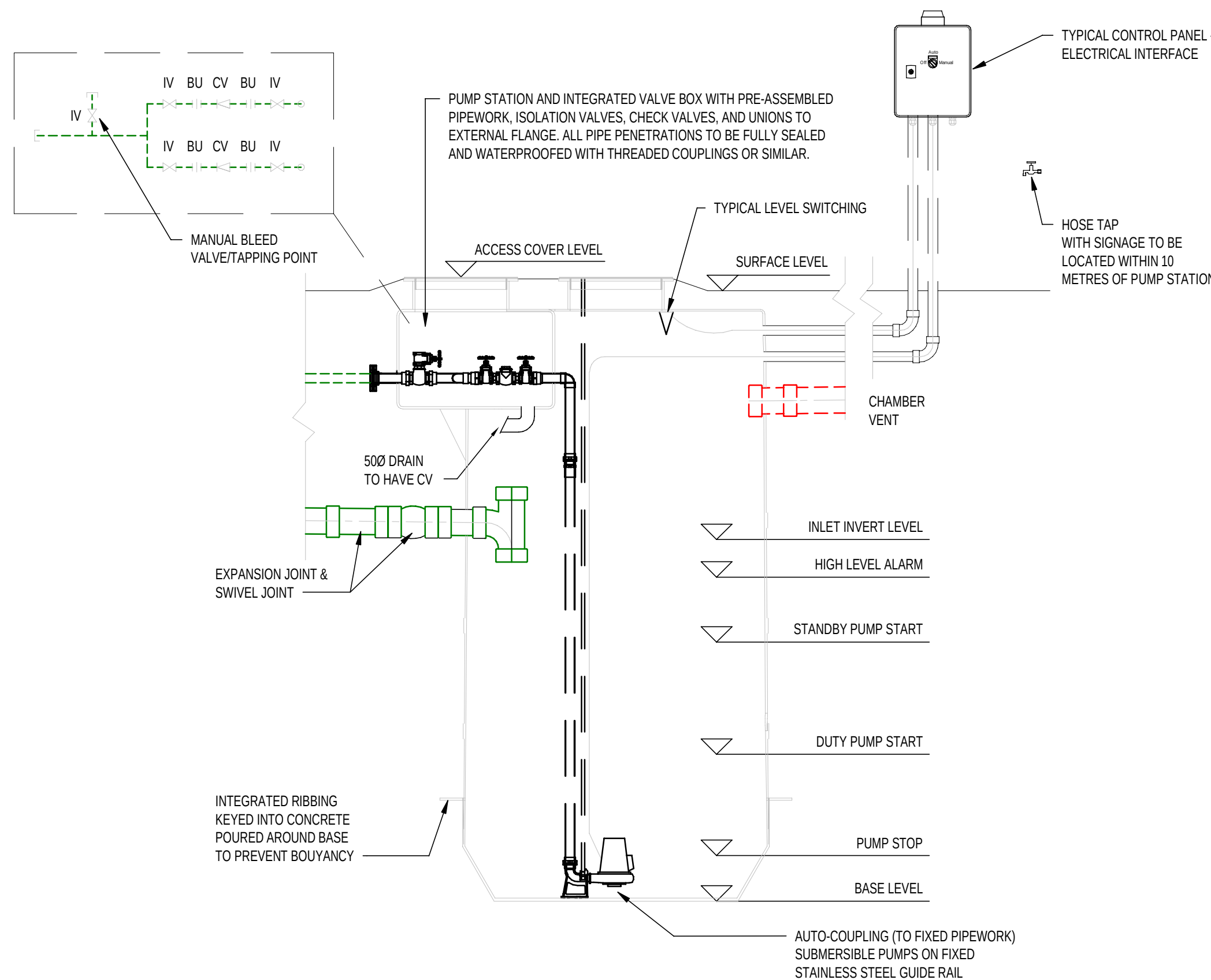
TMV DETAIL
N.T.S.



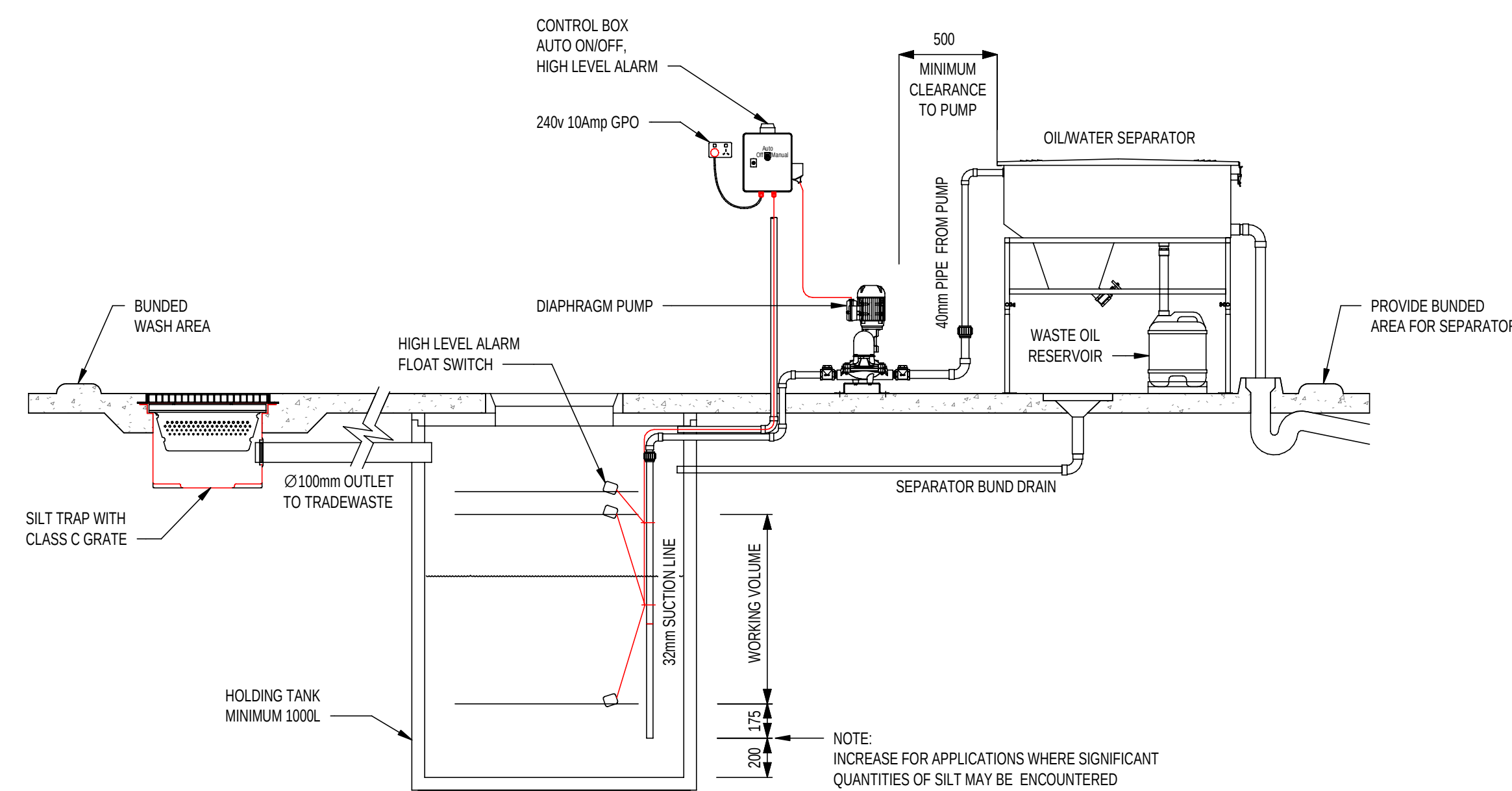
TRAFFICABLE IOS
N.T.S.



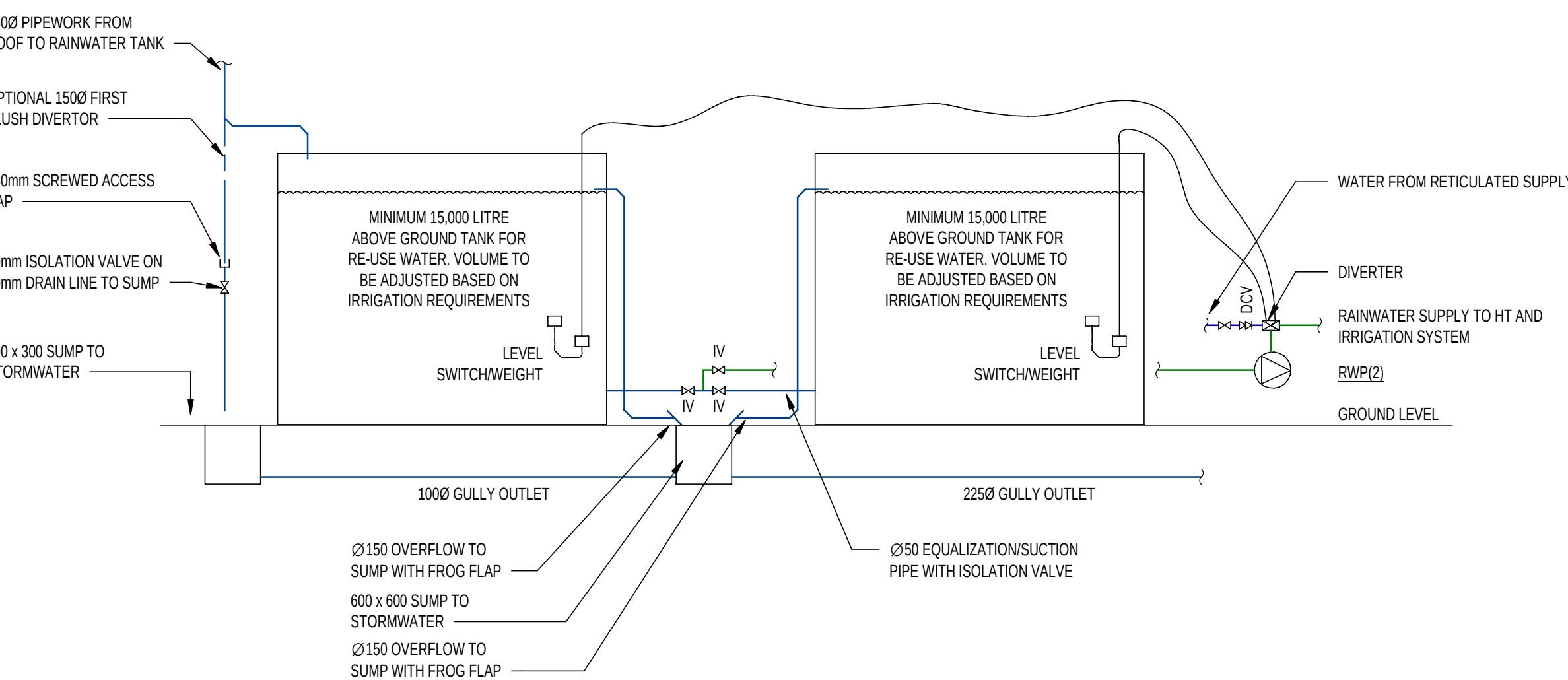
ZIP 4 IN 1 UNDER BENCH MODEL BOILING, CHILLED, HOT & COLD
N.T.S.



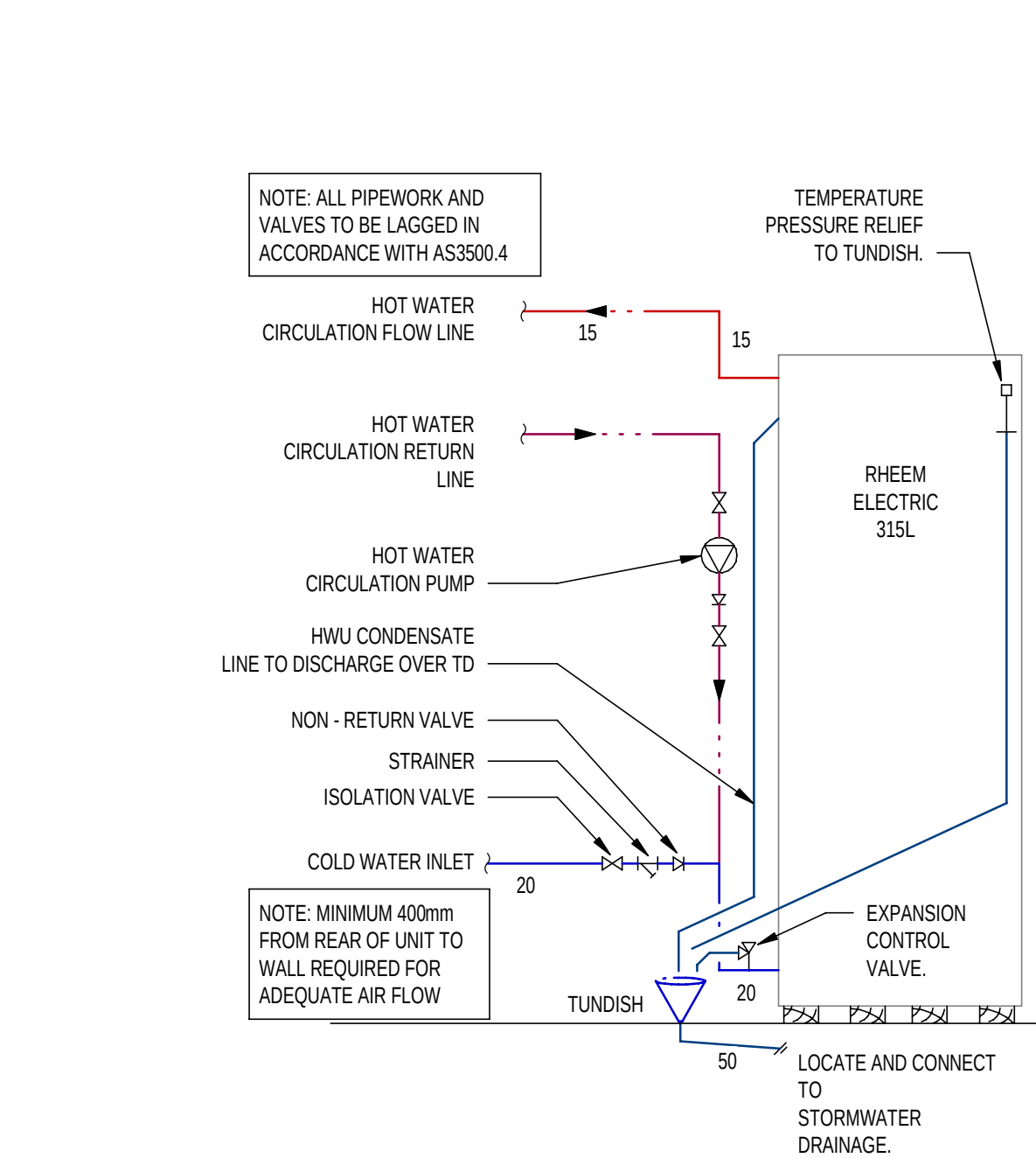
PACKAGED SANITARY DRAINAGE PUMP STATION DETAIL
N.T.S.



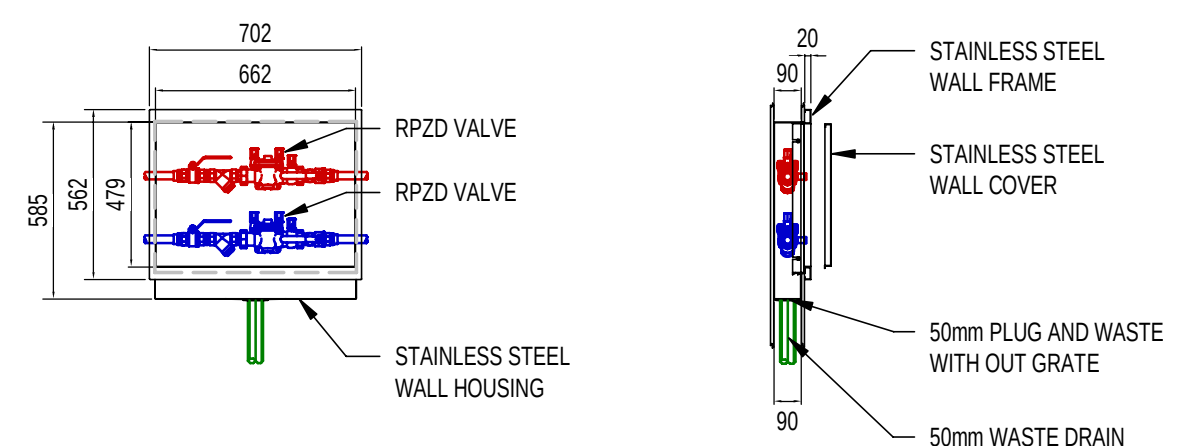
OIL/WATER SEPARATOR DETAIL
N.T.S.



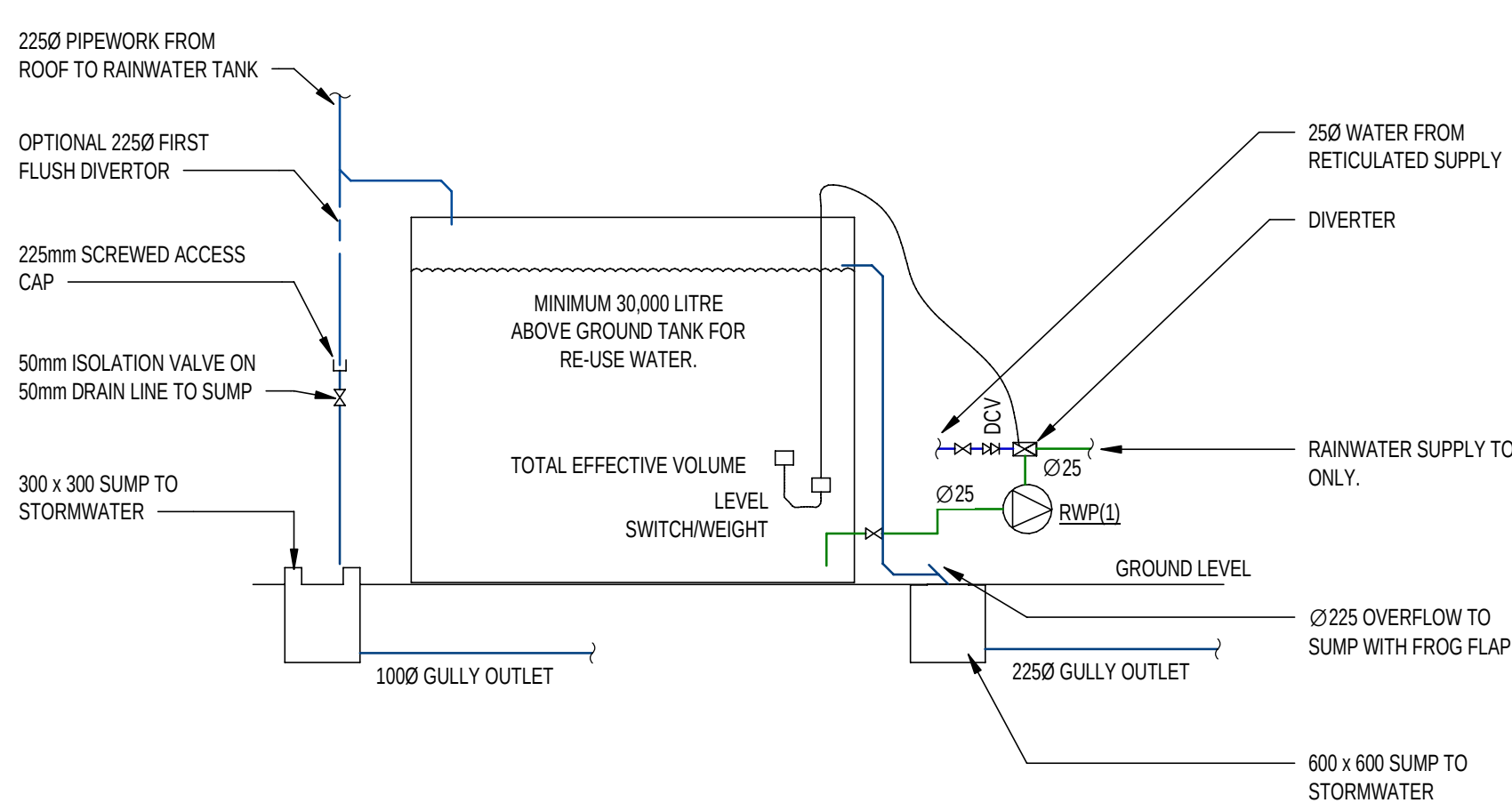
RAINWATER TANK DETAIL - HORTICULTURE BUILDING
N.T.S.



HOT WATER UNIT DETAIL
N.T.S.



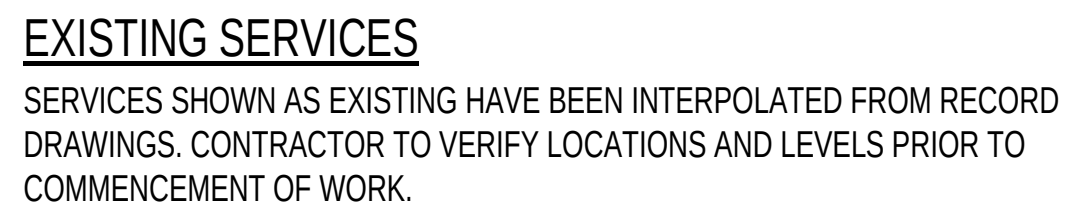
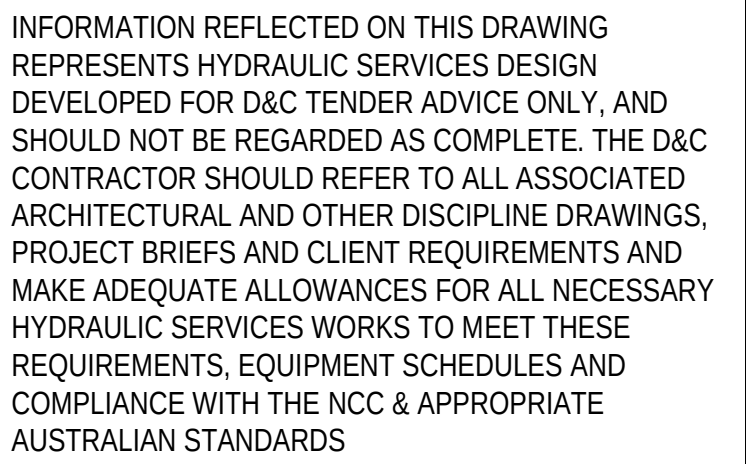
RPZD IN SS WALL BOX DETAIL
N.T.S.



RAINWATER TANK DETAIL - NEW TRADES BUILDING
N.T.S.

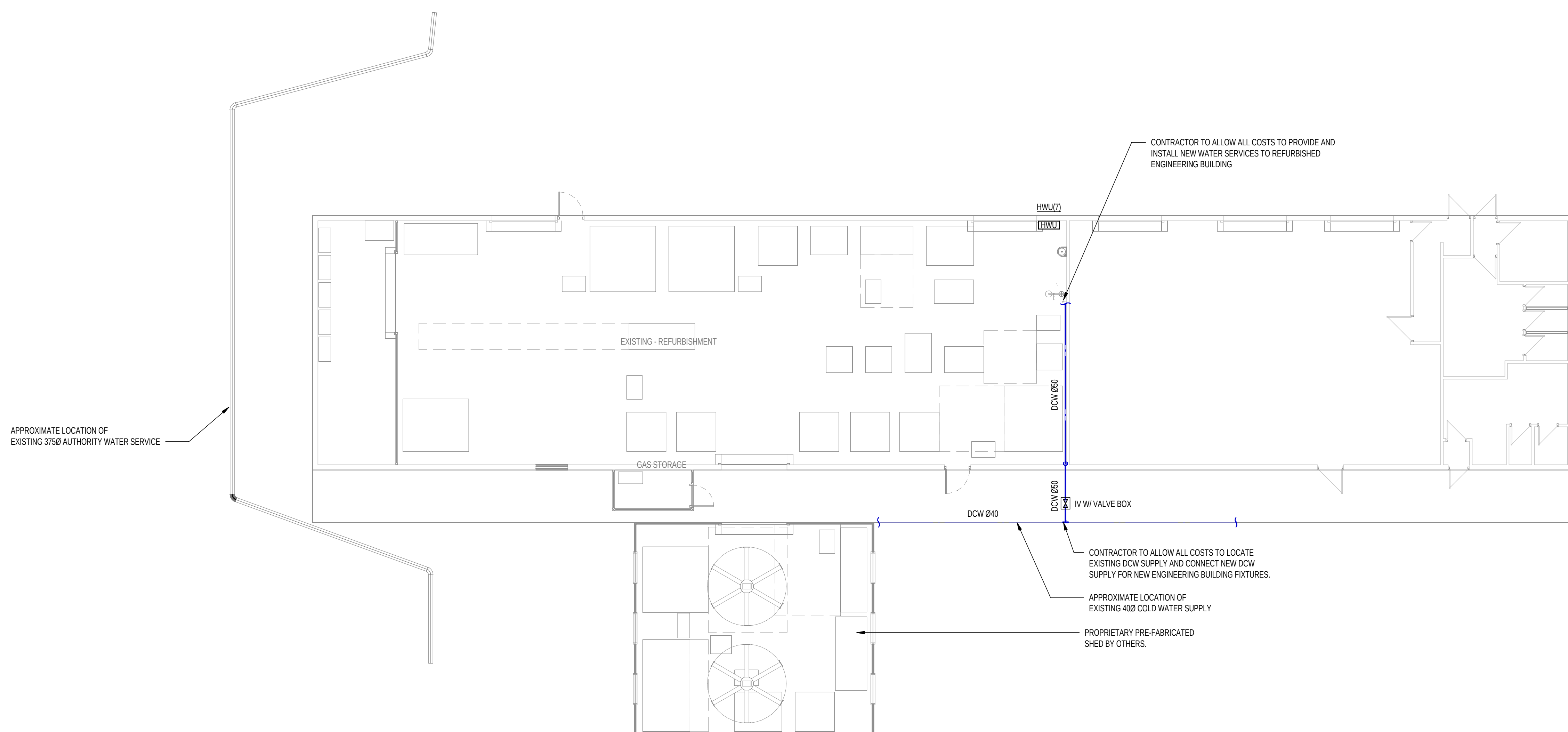
INFORMATION REFLECTED ON THIS DRAWING REPRESENTS HYDRAULIC SERVICES DESIGN DEVELOPED FOR D&C TENDER ADVISE ONLY, AND SHOULD NOT BE REGARDED AS COMPLETE. THE D&C CONTRACTOR SHOULD REFER TO ALL ASSOCIATED ARCHITECTURAL AND OTHER DISCIPLINE DRAWINGS, PROJECT BRIEFS AND CLIENT REQUIREMENTS AND MAKE ADEQUATE ALLOWANCES FOR ALL NECESSARY HYDRAULIC SERVICES WORKS TO MEET THESE REQUIREMENTS, EQUIPMENT SCHEDULES AND COMPLIANCE WITH THE NCC & APPROPRIATE AUSTRALIAN STANDARDS

A	12.02.2020	SH	ISSUED FOR D&C TENDER	AA	SM
REV	DATE	BY	DESCRIPTION	CHK	APP
DRAWING STATUS:					
DESIGN DEVELOPMENT					
 900 Ann Street, Level 12, Fortitude Valley 4006 Tel: +61-7-3854-6200 Fax: +61-7-3854-6500 http://www.wsp.com					
CLIENT:					
CQU					
ARCHITECT:					
THOMSON ADSETT / PEDDLE THORP					
PROJECT:					
CQU ROCKHAMPTON TAFE CONSOLIDATION					
TITLE:					
HYDRAULIC SERVICES TYPICAL DETAILS					
SCALE & NO.		CHECKED:	AA	APPROVED:	SM
PROJECT NUMBER:		DRAWN:	SH	DATE:	JAN 2020
DRAWING NO:		00-HY-DR-100-A00-00-02			REV:
					A
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- DRAINAGE NOTES:**
1. ALLOW ALL COSTS TO CONNECT SANITARY DRAINAGE TO ALL FUTURES AS REQUIRED.
 2. CO-ORDINATE ALL FLOOR WASTE LOCATIONS WITH ARCHITECT AND IN COMPLIANCE WITH THE NCC, AS/NZS3502.1, AS1428.1
 3. ALLOW FOR PROVISION OF AN OGS IN COMPLIANCE WITH AS/NZS3502.1 AND ALL COMPLETE DRAIN ACCESS IS ALL DOWN TO COLLECTING WATER PIPES.
 4. DOCUMENTATION MAY NOT INDICATE ALL VENTING REQUIREMENTS AND SHALL BE FINALISED DURING FURTHER DESIGN DEVELOPMENT. VENTING REQUIREMENTS TO BE IN ACCORDANCE WITH AS/NZS3502.1
 5. ALL PENETRATIONS OR CAST THROUGH CONCRETE STRUCTURE ARE TO BE APPROVED BY THE STRUCTURAL ENGINEER.
 6. ALLOW FOR DRAINAGE ARTICULATION AS REQUIRED AND SUBJECT TO THE FINAL SITE SLO CLASSIFICATION IN ACCORDANCE WITH AS2670
 7. VERIFY THAT ALL MECHANICAL DISCHARGE EQUIPMENT DO NOT EXCEED THE MAXIMUM ASSOCIATED WITH THE MECHANICAL MATERIALS MANUFACTURED TO PROVIDE DRAIN TEMPERING VALUE FOR DISCHARGE AS REQUIRED.
 8. CO-ORDINATE WITH ARCHITECT AND PROVIDE THROUGH PROTECTION AT ALL ENTRY POINTS TO THE BUILDING AS REQUIRED TO PREVENT THE INGRESS OF WIND DRIVEN RAIN.

INFORMATION REFLECTED ON THIS DRAWING
REPRESENTS HYDRAULIC SERVICES DESIGN
DEVELOPED FOR D&C TENDER ADVICE ONLY, AND
SHOULD NOT BE REGARDED AS COMPLETE. THE D&C
CONTRACTOR SHOULD REFER TO ALL ASSOCIATED
ARCHITECTURAL AND OTHER DISCIPLINE DRAWINGS,
PROJECT BRIEFS AND CLIENT REQUIREMENTS AND
MAKE ADEQUATE ALLOWANCES FOR ALL NECESSARY
HYDRAULIC SERVICES WORKS TO MEET THESE
REQUIREMENTS, EQUIPMENT SCHEDULES AND
COMPLIANCE WITH THE NCC & APPROPRIATE
AUSTRALIAN STANDARDS



EXISTING SERVICES

SERVICES SHOWN AS EXISTING HAVE BEEN INTERPOLATED FROM RECORD DRAWINGS. CONTRACTOR TO VERIFY LOCATIONS AND LEVELS PRIOR TO COMMENCEMENT OF WORK.

WATER NOTES:

1. ALLOW ALL COSTS TO RETICULATE ALL HOT AND COLD WATER FROM AREA/SLOPE TO TWO TIVOTS WITHIN THE FUTURE AREA.
2. ALL FUTURE AREAS TO HAVE MINN ISOLATION VALVES AT EACH CONNECTION FOR ALL MAINS WITH FLEXIBLE HOSES, OR ARE TO BE ISOLATED BY MEANS OF ISOLATING ONLY THE ROOM INDEPENDENTLY FROM THE REMAINDER OF THE BUILDING.
3. PROVIDE PRESSURE REDUCTION AS REQUIRED TO MAINTAIN FUTURE AREA PRESSURE WITHIN THE MAXIMUM AND MINIMUM ALLOWABLE PRESSURES SPECIFIED IN THE SPECIFICATION. ALL PRESSURE REDUCTION MAINLINES SHALL BE INSTALLED IN REDUNDANCY (INSTALLED IN PARALLEL) WITH UPSTREAM AND DOWNSTREAM ISOLATION OF EACH VALVE.
4. DOCUMENTATION MAY NOT INDICATE ALL LOCATIONS OF REQUIRED VALVING. FINAL REQUIREMENTS TO BE DETERMINED BY FURTHER DESIGN DEVELOPMENT.
5. WHEN REQUIRED, PROVIDE VALVING ACCESS PANELS TO MAINTAIN ACCESS TO ALL VALVING, PRESSURE AND FLOW MONITORING.
6. PROVIDE BACKFLOW PREVENTION DEVICES AS REQUIRED TO ENSURE THAT ZONE PROTECTION IS RETAINED IN ACCORDANCE WITH ASH208500.1 AND LOCAL AUTHORITY REQUIREMENTS.
7. CONTRACTOR TO REFER TO STRUCTURAL DRAWINGS FOR SPECIFIC CUTTING. ALLOW ALL COSTS FOR CUTTING AND REINSTATEMENT AS REQUIRED.
8. CONTRACTOR TO COORDINATE PROVISIONS WITH FINAL FLOOR LAYOUT PROVIDED BY ARCHITECT.

REV	DATE	BY	DESCRIPTION	CHK	APP
0	12.02.2020	SH	ISSUED FOR O&C TENDER	AA	SM
1	20.01.2020	SH	DRAFT O&C IDO ISSUE FOR REVIEW	AA	SM

DESIGN DEVELOPMENT

WSP

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<http://www.wsp.com>

CLIENT:

COU

ARCHITECT

THOMPSON ADSETT / PEDDIE THORP

PROJECT:

CQU ROCKHAMPTON TAFE CONSOLIDATION

TITLE:

HYDRAULIC SERVICES
ENGINEERING BUILDING - GROUND LEVEL - WATER
SERVICES LAYOUT

SCALE @ A0: 1:100	CHECKED: AA	APPROVED: SM
PROJECT NUMBER: PS229518	DRAWN: SH	DATE: JAN 2026

DRAWING No:	REV:
ENG-HY-DR-520-L00-00-00	B

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